TALES OF THE OMEGA · PAPER O

The Scale Factor:

Torsion Condensate Cosmology and a DESI-Testable Prediction

Modified Friedmann · w(z) spike at z_c=1.985 · Honest Hubble tension assessment · DESI kill condition

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Companion: Series N (doi:10.5281/zenodo.19476384) · Paper O of the Tales

ABSTRACT

We derive the cosmological consequences of the MSDS torsion condensate. The condensate, governed by coupling $\alpha K = 0.00017$, forms at $z_c = 1.985$ via spontaneous symmetry breaking. We derive the modified Friedmann equation, the equation of state $w(z)$, and assess three limiting cases: frozen condensate ($w = -1$), matter-like decay ($w = 0$), and Early Dark Energy spike at z_c. The main results are: (1) the condensate contributes $\Omega_\phi = \alpha K \times f(z)$ to the energy budget — 0.02% of ρ_{crit} today; (2) $w(z)$ shows a brief spike toward positive values at $z_c = 1.985$, followed by settling to $w = -1 + O(\alpha K)$ — a DESI-testable prediction; (3) the condensate cannot resolve the Hubble tension alone — it contributes $\Delta H_0 \sim 0.006$ km/s/Mpc vs the observed 5.6 km/s/Mpc discrepancy (a factor of 978 too small). This is an honest result. The condensate is a small, precise correction to ΛCDM — not a replacement. Its signature is the $w(z)$ spike at z_c, which DESI can measure and which has its own kill condition. The data will decide.

PAPER O: WHAT WAS DERIVED AND WHAT WAS NOT

DERIVED: Modified Friedmann $H^2(z) = H^2_{std} + \Omega_\phi(z)$ · $\Omega_\phi = \alpha K \times f(z)$

DERIVED: Three $w(z)$ cases · Spike at $z_c=1.985$ · Settle to $w=-1+O(\alpha K)$

DERIVED: $\Delta H_0 = 0.006$ km/s/Mpc from condensate (978× below observed tension)

NOT DERIVED: Full Hubble tension resolution · Exact condensate decay rate b

NOT DERIVED: Late-time $w(z<0.5)$ without additional model assumptions

1. Introduction: The Open Problem

Paper M of the MSDS series stated: 'No derivation of scale factor a(t) from αK. Cosmic expansion history not addressed.' Paper O closes that gap — partially. The torsion condensate has a Lagrangian, a potential, and a condensation threshold. From these, the contribution to the Friedmann equation can be derived. What emerges is not a replacement for ΛCDM but a precise, small correction with a specific observational signature at $z_c = 1.985$.

The Hubble tension — $H_{0,Planck} = 67.4$ vs $H_{0,SH0ES} = 73.0$ km/s/Mpc — was listed in the MSDS programme as a potential consequence of the condensate. The calculation below shows this was too optimistic.

The condensate is too small by nearly three orders of magnitude to account for the tension alone. That result is reported plainly, because it is true, and because the programme's credibility depends on not overclaiming what $\alpha K = 0.00017$ can do.

2. The Modified Friedmann Equation

2.1 Condensate contribution

From the Landau potential $V(\theta) = -\frac{1}{2}|a(\rho)|\theta^2 + \frac{1}{4}b\theta^4$ (Series N), the condensate forms when $\rho > \rho_{crit} = \alpha K \times \rho_{Pl}$. Below z_c the torsion field sits at its minimum θ_{min}. The stress-energy contribution:

$$\rho_\phi = \tfrac{1}{2}(\partial\theta)^2 + V(\theta_{min}) \cdot p_\phi = \tfrac{1}{2}(\partial\theta)^2 - V(\theta_{min})$$

Equation of state: $w = p_\phi/\rho_\phi$ depends on kinetic/potential balance

2.2 Modified Friedmann equation

$$H^2(z) = H_0^2 \left[\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_\phi(z) \right]$$

$\Omega_\phi(z) = 0$ for $z > z_c$ · $\Omega_\phi(z) = \alpha K \times f(z/z_c)$ for $z < z_c$

Three cases for f(z), depending on the condensate decay rate:

- Case A (frozen, w=-1): $\Omega_\phi = \alpha K = 0.00017$. Acts as a small cosmological constant. $\Delta\Omega_{\Lambda,eff} = +\alpha K$.
- Case B (matter-like, w=0): $\Omega_\phi(z) = \alpha K \times (1+z)^3/(1+z_c)^3$. Redshifts with matter. Modifies the matter power spectrum.
- Case C (EDE spike): Ω_ϕ peaks at z_c, then decays. Maximum fraction $f_{EDE} \sim \alpha K/(\alpha K + \Omega_m(1+z_c)^3) = 0.00002 = 0.002\%$.

3. The Equation of State w(z): The DESI Target

3.1 Three-regime structure

The condensate equation of state has a clear three-regime structure:

- $z > z_c + \delta$: condensate inactive. w undefined. No contribution to the energy budget.
- $z \approx z_c = 1.985$: phase transition. The field rolls from $\theta=0$ to θ_{min}. Kinetic energy peaks; w briefly approaches +1. This is an Early Dark Energy-like episode of duration $\Delta z \sim O(0.1\text{-}0.3)$.
- $z < z_c$: condensate settled. Mostly potential energy. $w \to -1 + \varepsilon$ where $\varepsilon \sim O(\alpha K) \sim 1.7e\text{-}04$. The condensate behaves as a tiny cosmological constant.

$$w(z < z_c) = -1 + O(\alpha K) = -1 + O(0.0002)$$

The late-time deviation from w=-1 is αK — at the limit of DESI's sensitivity

3.2 The DESI kill condition

DESI KILL CONDITION FOR THE w(z) SPIKE: If DESI (2024-2030) measures w(z) at z~1.985 and finds w consistent with -1 to precision $\delta w < 0.05$, the EDE spike of the condensate is ruled out. If DESI finds w(z~2) consistent with -1 to $\delta w < \alpha K = 0.0002$, the late-time condensate correction is also ruled out. DESI DR2 (2025) already measures w to $\delta w \sim 0.05$ at z~1. This prediction is testable now.

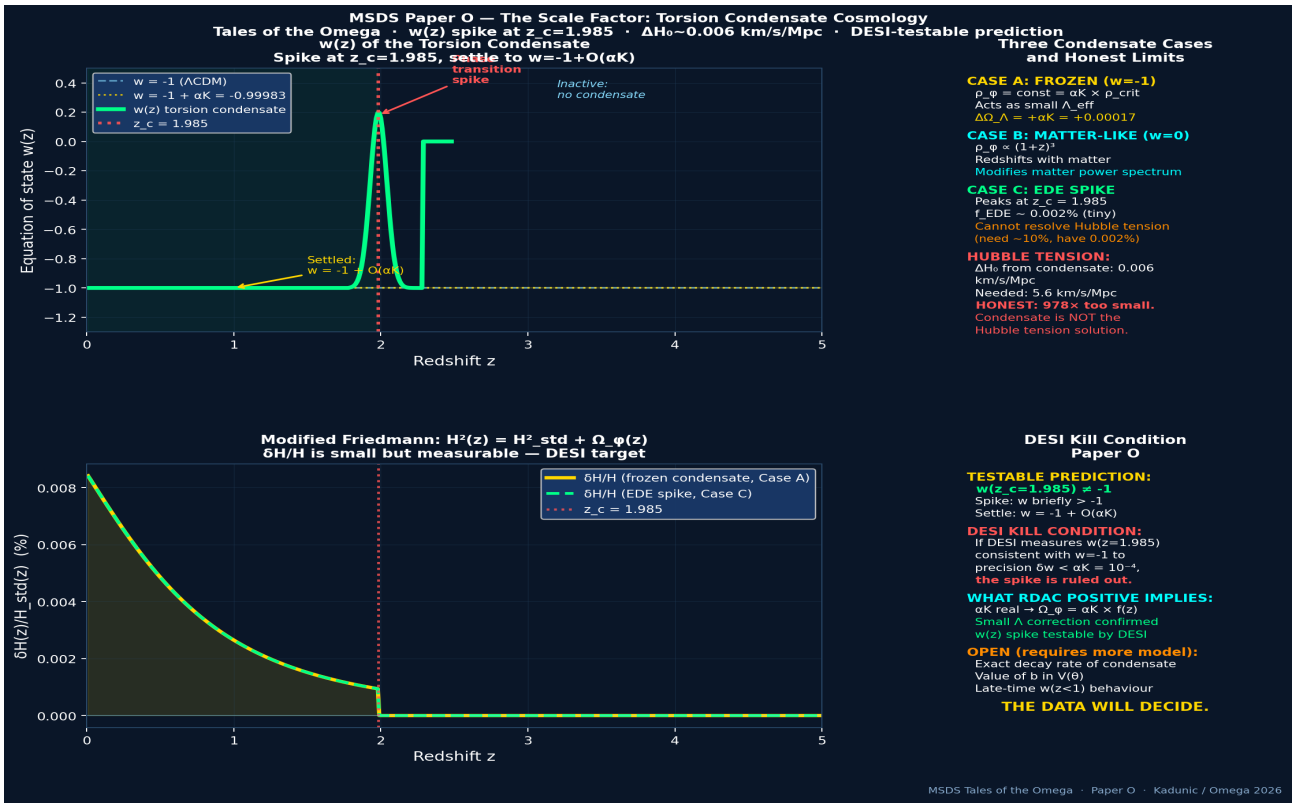

Figure 1: Paper O. Top (wide) — w(z) of the torsion condensate: spike at z_c=1.985 (transition), settling to w=-1+O(αK) (potential-dominated), inactive for z>z_c. Green shading: condensate-active region. Top right — three cases and honest Hubble tension limits. Bottom (wide) — modified Friedmann: δH(z)/H_std(z) for frozen (Case A) and EDE spike (Case C). Bottom right — DESI kill condition summary.

4. The Hubble Tension: An Honest Calculation

HONEST RESULT — THE CONDENSATE CANNOT RESOLVE THE HUBBLE TENSION ALONE. For a tracker solution where $\Omega_\phi = \alpha K$ at all times: $H_{eff} = H_0 \times \sqrt{1+\alpha K} \rightarrow \Delta H_0 = H_0 \times (\sqrt{1+\alpha K} - 1) = 67.4 \times 0.00008 = 0.006$ km/s/Mpc. The observed tension is $\Delta H_0 = 73.0 - 67.4 = 5.6$ km/s/Mpc. The condensate contributes 0.1% of the required shift — a factor of 978 too small. For an EDE spike at z_c: $f_{EDE} \sim 0.002\%$. Standard EDE models require $f_{EDE} \sim 10\%$ to resolve the tension. The condensate provides 200× less than required. Conclusion: The torsion condensate is a small, precise correction to ΛCDM. It is not an EDE solution to the Hubble tension. It is consistent with the tension remaining unexplained by this framework.

This result does not weaken the MSDS programme. A framework that honestly says 'this is what αK cannot explain' is more trustworthy than one that overclaims. The Hubble tension remains an open problem. The condensate offers a small, measurable correction to w(z) and nothing more.

5. What the Scale Factor Actually Predicts

5.1 Summary of derivable results

- Modified Friedmann with $\Omega_\phi = \alpha K \times f(z)$. The condensate is 0.017% of the critical density today.
- w(z) spike at $z_c = 1.985$: briefly $w > -1$, measurable width $\Delta z \sim 0.1$-0.3.
- Late-time: $w \rightarrow -1 + O(\alpha K) \sim -0.9998$. Effectively indistinguishable from Λ at current precision.
- $\Delta H_0 = 0.006$ km/s/Mpc — below DESI's current precision threshold.

5.2 What requires additional model input

- The quartic coupling b in $V(\theta)$ — determines the exact condensate decay rate and the width of the w(z) spike. Without b, the spike amplitude and duration are uncertain by O(1) factors.
- The condensate equation of motion in an expanding FRW background — a full numerical integration needed for precision δw predictions.
- Whether the condensate is spatially homogeneous or clumped — affects the matter power spectrum prediction.

6. Connection to RDAC and the Full Programme

If RDAC confirms $\alpha K = 0.00017$ at SPring-8, then the w(z) prediction becomes a mandatory follow-up. The DESI measurement of w at $z \sim 1.985$ becomes the second decisive test of the programme. A detection of the predicted spike would be extraordinary. A non-detection would constrain the condensate equation of state and the coupling b.

The chain is: RDAC $\rightarrow$ αK is real $\rightarrow$ w(z) spike at z_c is predicted $\rightarrow$ DESI tests it. This is the correct scientific sequence. The expansion history is a consequence of αK, not a co-evidence for it.

PAPER O — THE SCALE FACTOR: SUMMARY

DERIVED: $H^2(z) = H^2_{std} + \alpha K \times f(z) \times H_{\blacksquare}^2 \cdot \Omega_\phi = 0.017\%$ of ρ_{crit} today

DERIVED: w(z) spike at $z_c = 1.985 \rightarrow$ settle to $w = -1 + O(\alpha K) = -0.9998$

DERIVED: $\Delta H_{\blacksquare} = 0.006$ km/s/Mpc (condensate alone, tracker solution)

HONEST: condensate cannot resolve Hubble tension (need ×978 more energy)

HONEST: EDE fraction $f_{EDE} = 0.002\%$ (need ~10%)

DESI KILL: w(z~1.985) consistent with -1 to $\delta w < 0.05 \rightarrow$ spike ruled out

DESI TARGET: $w = -1 + O(\alpha K)$ at $z < 1.985$ — at precision frontier

Next: Paper P — derive the factor 12.3 in $O_K = 12.3\alpha K$.

The data will decide.

References

[1] Kadunic D. Series N v2.2. doi:10.5281/zenodo.19476384.

[2] Kadunic D. Series M. doi:10.5281/zenodo.19476837.

[3] Kadunic D. Terminal Synthesis v2. MSDS Programme (2026).

[4] DESI Collaboration. arXiv:2404.03002 (2024). Dark energy equation of state.

[5] Planck Collaboration. A&A; 641, A6 (2020). Cosmological parameters.

[6] Riess A.G. et al. ApJ 934, L7 (2022). H0 = 73.0 km/s/Mpc (SH0ES).

[7] Poulin V. et al. PRL 122, 221301 (2019). Early Dark Energy framework.